

Security Essentials

Computing and Mathematics

May, 2026

Short Title:	Security Essentials	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Forensics and Security	
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Credits	5	Level: Intermediate

Description of Module / Aims

This module provides an introduction to IT security. Topics covered include various security threats and vulnerabilities and the services available to address these threats. The importance of security policies and procedures within an organisation is explained. Cryptographic techniques that underpin many security mechanisms are covered. Finally, best practice in secure software development, with an emphasis on web applications, is also explored.

Programmes

COMP-0533 Diploma in Computing with Security and Forensics (WD_BCSEC_SP)

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Indicative Content

- Introduction to computer security: security threats; attack methods; common vulnerabilities; security services
- Security management: security assessment; security design; policy
- Cryptography: symmetric encryption; modern encryption; block vs stream ciphers; public-key cryptography
- Message authentication and integrity; digital signatures; digital certificates
- Digital certificates and TLS (transport layer security)
- Web application security; OWASP Top Ten
- Secure programming: software vulnerabilities; coding standards

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Categorise the various security threats and attack methods to which an organisation may be susceptible.
2. Demonstrate practical knowledge of the security technologies that must be implemented to protect an organisation.
3. Appraise the role of cryptography in computer security, including its benefits and limitations.
4. Demonstrate the use of encryption software for data confidentiality and authentication purposes.
5. Justify the importance of security policies and procedures within an organisation.
6. Examine the specific security concerns pertinent when developing web applications.
7. Recommend how to build security measures into the software development process.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and practical classes.
- The lectures will be used to introduce new topics and their related concepts.
- The practical element allows the student to put into practice the theoretical concepts covered in the lectures.

- The practical element will be largely based on laboratory exercises involving the use of cryptographic software to encrypt data and configure secure communications (e.g. using TLS) as well as web application security tools.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,3,5,6,7
Continuous Assessment	50%	
<i>Practical</i>	50%	2,4,6

Assessment Criteria

- <40%:** Unable to interpret and describe key concepts of computer security.
- 40%-49%:** Able to interpret, describe and apply in some specific situations key concepts of key concepts of computer security.
- 50%-59%:** Able to discuss key concepts of computer security and apply in a variety of diverse ways.
- 60%-69%:** Able to experiment with security tools and configure them to demonstrate a detailed appreciation of security essentials.
- 70%-100%:** All the above to an excellent level. Able to suggest solutions for a range of both complex problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- "Open Web Application Security Project." <https://www.owasp.org>

Supplementary Material

- "Computer Emergency Response Team." <https://www.cert.org>
- "Security Focus." <http://securityfocus.com/>
- "The SANS Institute." <https://www.sans.org>
- Gollmann, D. _Computer Security_. 3rd ed. NY: Wiley, 2011.
- McGraw, G. _Software Security: Building Security In_. NY: Addison-Wesley, 2006.
- Stallings, W. and L. Brown. _Computer Security: Principles and Practices_. 3rd ed. NY: Pearson, 2014.

Requested Resources

- COMPUTER LAB: BYOD Lab